

Primary OSCC study: UOP + STA
48 patients / 142 ROIs / 498,238 annotated cells / 39 markers
Context: 2 SCCHN samples and 5 descriptive cross-cancer cohorts



Frozen virtues-sp32: checkpoint hash and marker identities recorded
Blur → uq0.99 clipping → log1p → pooled image z-score
Global normalization is transductive; cohort lists do not establish sample exclusion



Task 1
Cell phenotyping



Task 2
Channel reconstruction



Task 3
Token clustering



Matched patient folds
Token / intensity / fusion

Labels derived from
intensity measurements



72 windows / 36 patients
39 channels individually
masked

Spatial null + ridge
LOWO / LOPO + paired CI



UOP fitting; STA assignment
Multi-seed stability

Patient-paired Ki-67
Exploratory recurrence



Outputs: per-marker differences, uncertainty and claim-to-evidence map
Spatial alignment ≠ incremental performance ≠ patient-disjoint validation
Unstable partition and negative Ki-67 findings retained
Reusable protocol and executable result summaries